Ansik Aryan Samal

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PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, 2025, CGPA 7.83) and AI Engineer intern at Clyptus Software Solutions, Hyderabad. Strong in Python, SQL and C++ with hands-on experience in software development, data science and machine learning projects. Built a job authenticity tool with 95% accuracy on unseen data, and secure REST APIs with role-based access control, and took a client product from requirements to launch. Used generative AI and automation (prompt engineering, RAG, n8n) to streamline workflows; currently completing a Data Science and Artificial Intelligence program with IIT Roorkee and Masai School and working on Text-to-SQL analytics copilot with query safety guardrails.

TECHNICAL SKILLS

- **Operating Systems, Networking and Troubleshooting:** Linux/Unix, operating systems fundamentals, computer networks fundamentals, debugging, root cause analysis (RCA), unit testing and creating PWAs (Progressive Web Applications).
- **Programming and Scripting:** Python, SQL, Bash and shell scripting, JavaScript, TypeScript, Java, C++, C
- **Databases and SQL:** PostgreSQL, MySQL, MongoDB, Supabase, Redis, DuckDB; data modelling, query optimization, SQL for data retrieval, cleaning and reporting
- **Automation and DevOps Concepts:** automation and workflow automation, n8n, DevOps concepts, CI/CD concepts (automated testing and releases), Docker, Git, GitHub, Postman; deployment on Railway, Render, Vercel and Cloudflare (custom domains)
- **Cloud Fundamentals:** AWS fundamentals, cloud deployment basics
- **Generative AI and AI Productivity Tools:** prompt engineering (one-shot, few-shot, chain-of-thought, role-based prompting), retrieval-augmented generation (RAG), embeddings and vector search, LLM evaluation, agentic workflows, Model Context Protocol (MCP); Claude Code, Cursor, Codex, ChatGPT, Lovable, Emergent, Replit, Perplexity
- **AI and Machine Learning:** Scikit-learn, regression, classification, feature engineering, Hugging Face inference models
- **Data Science and Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, exploratory data analysis (EDA), data cleaning, statistical analysis, trend analysis, Power BI, Tableau, Microsoft Excel, dashboards and metrics reporting
- **Back End and APIs:** Python (Flask, FastAPI), Node.js, Express.js, RESTful APIs, WebSockets, authentication (Passport.js, bcrypt, JWT), role-based access control (RBAC)
- **Front End:** React.js, HTML, CSS, Tailwind CSS, responsive design
- **Computer Science Fundamentals:** Data Structures and Algorithms, object-oriented programming (OOP), SOLID principles, software development life cycle (SDLC)
- **Currently Working On:** Advanced RAG, Advanced Machine Learning, model fine-tuning, Generative AI and cloud fundamentals; also learning CI/CD, low-level design (LLD) and caching
- **Professional Skills:** analytical thinking, problem solving, technical documentation, clear written and verbal communication, collaboration, customer-focused mindset, adaptability, leadership

PROJECTS

Text-to-SQL Analytics Copilot with Guardrails | Python, sqlglot, FastAPI, DuckDB/PostgreSQL, Chroma/pgvector, Plotly, Streamlit

- Built a copilot that lets business users ask data questions in plain English and returns a result table, an auto-generated chart and a plain-English explanation of the query, removing the wait for an analyst on simple questions.
- Retrieves the relevant database schema, generates SQL, validates it and runs it read-only, so wrong joins, invented columns and destructive queries never reach the database.
- Parses every query into an abstract syntax tree (AST) with sqlglot and enforces safety rules in code instead of trusting the language model.
- Added a self-correction loop that feeds database error messages back to the model, plus a query bank of verified question-to-SQL pairs that serves as few-shot examples and regression tests.
- Evaluated against a gold set of 100 question-to-expected-result pairs, measuring execution accuracy, safety violations (target: zero) and median latency.

Job and Company Verification Tool | Python, Flask, React.js, TypeScript, Supabase, Tailwind CSS, TanStack, Firecrawl

- Engineered a tool that checks a single job link, gathers company reviews and online presence data through Firecrawl, and flags warning signs in the posting.
- Assigns each job an authenticity score from 0 to 100, reaching 95% accuracy on unseen data.
- Delivered the first version with a Flask REST API, a nine-signal weighted scoring system, a unit test suite and a 39-page PDF report (ReportLab), deployed on Render, Railway and Vercel.

- Introduced a bulk self-training feature that trains the model on batches of 30-40 company links, with predictions verified through Firecrawl.
- Rebuilt the app around an in-house machine learning model that classifies job listings, after applying feature engineering learned in the Data Science course.

Finance Data Processing Backend | Node.js, Express.js, PostgreSQL

- Developed secure REST APIs for user management, transactions and dashboard reports.
- Implemented role-based access control with three permission levels: Viewer, Analyst and Admin.
- Created endpoints that report income, expenses, balances and spending trends.
- Optimized database schemas and queries so dashboard summaries load quickly in real time.

Stock Screener SaaS Web App (Client Project) | React.js, Tailwind CSS, Node.js, Flask, Axios, MongoDB

- Translated a client's requirements into a customizable stock screener covering 5,300+ NSE and BSE stocks, with live prices, watchlists, portfolio tracking and real-time analysis.
- Defined Free, Pro ($20/month) and Premium ($40/month) plans with discounted yearly rates, and tied features such as stock search, number of watchlists and automatic price refresh to each plan.
- Determined the pricing and payment integration approach and wrote the project documentation.
- Launched on a custom domain through Cloudflare, reaching about 10,000 views in 3-4 months.

WORK EXPERIENCE

Clyptus Software Solutions Pvt Ltd, Hyderabad | AI Engineer Intern | Present

NxtWave Disruptive Technologies | Associate Aptitude Instructor | 24th August 2026 - 13th October 2026

- Trained to teach quantitative aptitude and math modules for computer science learners accurately and clearly, following company guidelines and ethics.
- Worked in a corporate setting with limited resources, staying punctual and raising issues early with the team.

Edunet Foundation (IBM SkillBuild) | Cybersecurity Intern | 15th January 2025 - 26th February 2025

- Learned core cybersecurity concepts, network security and threat mitigation techniques.
- Studied steganography methods for hiding and retrieving data within images using bit-level manipulation.
- Built an interactive GUI application that encodes and decodes hidden data within image files.

EDUCATION

Institute of Aeronautical Engineering (JNTUH), Dundigal, Hyderabad | B.Tech, Computer Science and Engineering | 2021 - 2025 | CGPA: 7.83 (out of 10)

Trividyaa Junior College, Hyderabad | Intermediate (MPC) | 2019 - 2021 | 98%

St. Michael's School, Old Alwal, Hyderabad | Secondary School | 2018 - 2019 | 94%

CERTIFICATIONS

- Data Science and Artificial Intelligence | IIT Roorkee and Masai School | Ongoing, 2026
- Python, SQL and Problem Solving | HackerRank
- Web Development (WD101) and Node Development (WD201) | Pupilfirst (AICTE-based curriculum)
- SEBI Investor Awareness | SEBI/NISM | 2025
- In preparation: Oracle AI, Microsoft Fabric Analytics Engineer, Anthropic, and Google AI and Cloud certifications

ACHIEVEMENTS AND LEADERSHIP

- Selected through the University of Hyderabad (UoH) entrance examination; ranked 162nd in the Common Rank List (CRL) and 4th in the Armed Personnel Category.
- Event Manager, Entrepreneurship Cell (E-Cell), IARE, for the National Entrepreneurship Challenge (NEC) E-Summit 2024 at IIT Bombay; the team's startup pitch ranked 4th in India.
- Pitched a job-matching app for daily-wage and off-season workers, with regional language support, salary negotiation help and planned tie-ups with government welfare schemes for fair wages.
- Scored 95.2 percentile in JEE Mains 2021.
- Qualified for the final mock interview round in LEEE 2023.

LANGUAGES

English (Professional), Hindi (Native), Telugu (Conversational), Odia (Conversational)